

SPECIAL NOTE: To receive full credit, you must show all of your work and write your final answers in the spaces provided in this booklet.

Standards of Conduct- Academic Integrity

As a student and member of the UIC community, you are expected to adhere to the Community Standards of integrity, accountability, and respect in all your academic endeavors.

Cheating on Examinations: Either intentionally using or attempting to use unauthorized materials, information, people, artificial intelligence, or study aids; providing to, or receiving from another person, any kind of unauthorized assistance on any examination.

Please note a new Chemical Engineering department policy: Cell phone, electronic eyeglasses or other type of electronic device usage are strictly prohibited during examinations.

The above behaviors will not be tolerated and will result at a minimum in a zero on the exam or assignment. Other possible actions include an "F" on the final grade, investigation, and disciplinary action by the Office of the Dean of Students and in egregious cases, expulsion from UIC.

1	50
2	50
T	100

- (1) (50 pts.) Consider the block flow diagram below. Notice that the streams are labeled 1 through 5. Species molar flow rates are given by the species symbol with the stream number as a subscript. The fresh feed contains only species A. This stream is mixed with a recycle stream containing A and P where P is the reaction product. The mixture enters a reactor in which one reaction takes place:

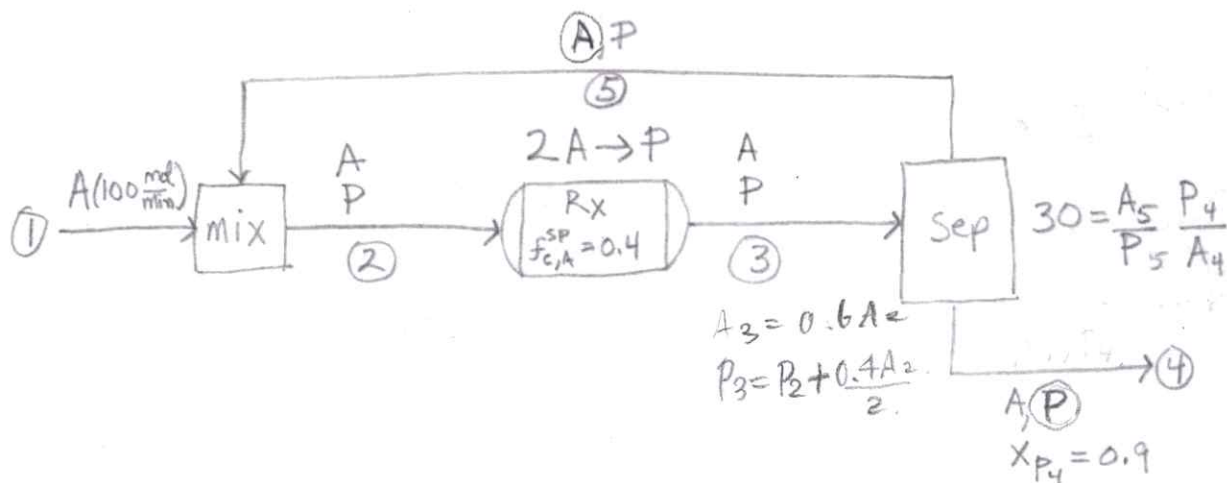


2 mol $\rightarrow$ 1 mol

The reactor converts 40% of A entering the reactor. The outlet stream from the reactor goes to a separator. Both streams leaving the separator are in equilibrium containing some A and P such that:

$$\alpha_{AB} = \frac{A_5 P_4}{P_5 A_4}$$

where the separation factor $\alpha_{AB} = 30$. Stream 5 is the stream enriched in A and is recycled back to the reactor. Stream 4 is enriched in the product P such that it is 90 mol% P. The fresh feed has species flow rates of $A_1 = 100$ mol A/min.



- (a) Use a degree of freedom analysis to determine whether the problem as stated is well posed. (5 pts.)

DoF

4a 9 : the number of components.

4b 1 : 1 reaction

5a 1 : $A_1 = 100$ mol/min

5b 1 : $x_{P,4} = 0.9$

5c 2 : $f_{c,A}^{sp}$ & α_{AB}

5d $(2+2+2) = 6$ components in each process (mix, reactor & separator)

10

$$DoF = 10 - 10 = 0$$

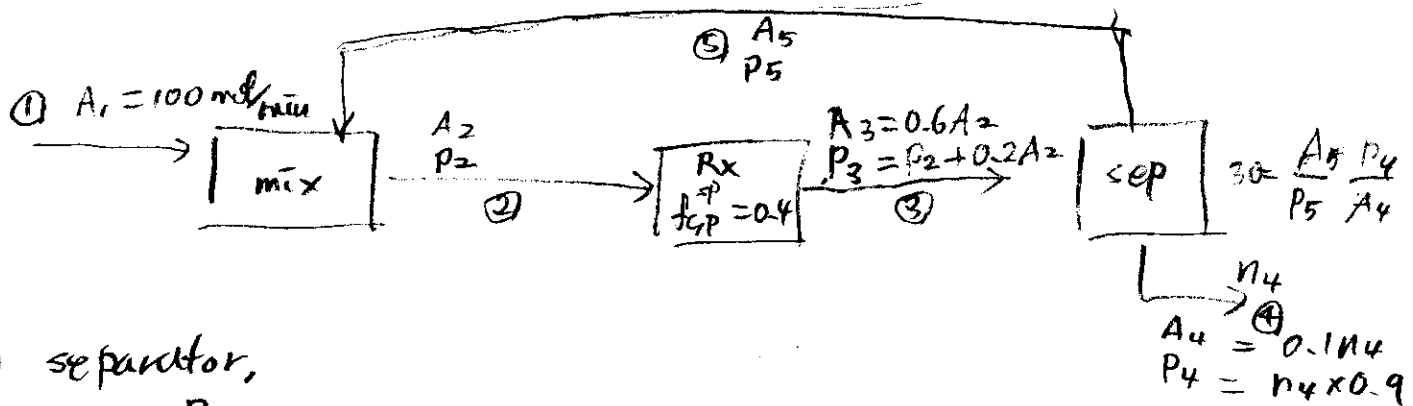
DoF = 0

(Problem 1 Continued)

- (b) Find all of the species molar flow rates not given in the problem statement. Fill in the table below with the appropriate species flow rates in mole/min to receive full credit. (40 pts.)

fraction separator: $0.4 \Rightarrow A_3 = A_2 - 0.4 A_2 = 0.6 A_2$

$$P_3 = P_2 + \frac{\gamma_P}{\gamma_A} 0.4 A_2 = P_2 + 0.2 A_2$$



In separator,

$$30 = \frac{A_5}{P_5} \frac{P_4}{A_4} = \frac{A_5}{P_5} \frac{0.9 n_4}{0.1 n_4} = 9 \frac{A_5}{P_5} \quad A_5 = \frac{30}{9} P_5$$

$$A_3 = A_4 + A_5 \Rightarrow 0.6 A_2 = 0.1 n_4 + \frac{30}{9} P_5 \quad (1)$$

$$P_3 = P_4 + P_5 \Rightarrow P_2 + 0.2 A_2 = 0.9 n_4 + P_5 \quad (2)$$

In mixer,

$$A_1 + A_5 = A_2 \Rightarrow 100 \text{ mol/min} + \frac{30}{9} P_5 = A_2 \quad (3)$$

$$P_5 = P_2 \quad (4)$$

4 unknowns & 4 equations

$$P_2 + 0.2 A_2 = 0.9 n_4 + P_2 \quad (\text{due to equation (4)})$$

$$n_4 = \frac{0.2}{0.9} A_2$$

$$\text{equation (1): } 0.6 A_2 - 0.1 \left(\frac{0.2}{0.9} A_2 \right) = \frac{30}{9} P_5$$

$$\text{equation (3): } A_2 - \left(0.6 A_2 - 0.1 \left(\frac{0.2}{0.9} A_2 \right) \right) = 100 \text{ mol/min}$$

$$0.42 A_2 = 100 \text{ mol/min}$$

$$A_2 = \frac{100 \text{ mol/min}}{0.42} = 238.1 \text{ mol/min}$$

$$n_4 = \frac{0.2}{0.9} (238.1 \text{ mol/min}) = 52.9 \text{ mol/min}$$

(Problem 1 Continued)

$$A_4 = (0.1) n_4 = (0.1) (52.9 \text{ mol/min}) = 5.29 \text{ mol/min}$$

$$P_4 = (0.9) n_4 = (0.9) (52.9 \text{ mol/min}) = 47.61 \text{ mol/min}$$

$$\frac{30}{9} P_5 = 0.6 A_4 - 0.1 \left(\frac{0.2}{0.9} A_2 \right) = 137.6 \text{ mol/min}$$

$$P_5 = \frac{9}{30} (137.6 \text{ mol/min}) = 41.28 \text{ mol/min}$$

$$P_2 = P_5 = 41.28 \text{ mol/min}$$

$$A_3 = 0.6 A_2 = 142.86 \text{ mol/min}$$

$$P_3 = P_2 + 0.2 A_2 = 41.28 \text{ mol/min} + 0.2 (238.1 \text{ mol/min}) \\ = 88.9 \text{ mol/min}$$

$$A_5 = \frac{30}{9} P_5 = 137.6 \text{ mol/min}$$

40

Species↓/Steam→	1	2	3	4	5
A	100	238.1	142.86	5.29	137.6
P	----	41.28	88.9	47.61	41.28

(Problem 1 Continued)

(c) Calculate the overall fractional conversion based on species A (5 pts.)

$$f_{C/A}^{ov} = \frac{A_{in} - A_{out}}{A_{in}} = \frac{A_1 - A_4}{A_1} = \frac{100 - 5.29}{100} \times 100(\%)$$
$$= 94.71\%$$

$$f_{C,A}^{ov} = \underline{94.71\%}$$

5

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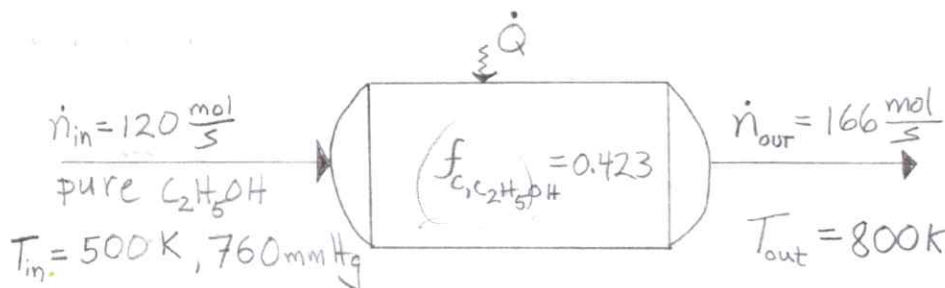
(2) (50 pts.) Acetaldehyde (CH_3CHO) is produced by dehydrogenation of ethanol ($\text{C}_2\text{H}_5\text{OH}$).



An undesired side reaction produces ethyl acetate ($\text{CH}_3\text{COOC}_2\text{H}_5$)



This process is carried out in a single reactor operating at steady state. The reactor has a feed rate of 120 mol/s of pure ethanol at 500K and 760 mm Hg. The reactor total product flow rate is 166 mol/s at 800K. The reactor converts 42.3% of the ethanol.



Data: The heat capacity in J/mol°C is given by $C_p = A + Bt$ where $t = T(\text{K})/1000$. The table below gives the constants A and B .

Species	A	B
$\text{C}_2\text{H}_5\text{OH}$	32.3	120
CH_3CHO	30.7	88.2
$\text{CH}_3\text{COOC}_2\text{H}_5$	60.9	191
H_2	28.4	1.54

- (a) Calculate all the species molar flow rates for the product stream in mol/s and write them into the Input/Output Table (see next page). Also, calculate the two extents of reaction, ξ_1 and ξ_2 . (20 pts.)

$$f_{\text{C}_2\text{H}_5\text{OH}} = 0.423 \Rightarrow \dot{n}_{\text{out}, \text{E}} = (1 - 0.423) \dot{n}_{\text{in}, \text{E}} = 0.577 \dot{n}_{\text{in}, \text{E}} = 69.24 \text{ mol/s}$$

$$\dot{n}_{\text{out}, \text{E}} = \dot{n}_{\text{in}, \text{E}} - \xi_1 - 2\xi_2 \quad \left| \quad \dot{n}_{\text{in}, \text{E}} = 120 \text{ mol/s} \right.$$

$$\dot{n}_{\text{out}, \text{A}} = \dot{n}_{\text{in}, \text{A}}^0 + \xi_1$$

$$\dot{n}_{\text{out}, \text{EA}} = \dot{n}_{\text{in}, \text{EA}}^0 + \xi_2$$

$$\dot{n}_{\text{out}, \text{H}} = \dot{n}_{\text{in}, \text{H}}^0 + \xi_1 + 2\xi_2$$

$$\dot{n}_{\text{out}} = \dot{n}_{\text{out}, \text{E}} + \dot{n}_{\text{out}, \text{A}} + \dot{n}_{\text{out}, \text{EA}} + \dot{n}_{\text{out}, \text{H}}$$

$$= \xi_1 + \xi_2 + \dot{n}_{\text{in}, \text{E}}$$

$$166 \text{ mol/s} = \xi_1 + \xi_2 + 120 \text{ mol/s}$$

$$\xi_1 + \xi_2 = 46 \text{ mol/s}$$

$$\xi_1 + 2\xi_2 = \dot{n}_{\text{in}, \text{E}} - 0.577 \dot{n}_{\text{in}, \text{E}} = 0.423 \times 120 \text{ mol/s} = 50.76 \text{ mol/s}$$

$$\begin{aligned} &\xi_1 + \xi_2 = 46 \text{ mol/s} \\ &\xi_1 + 2\xi_2 = 50.76 \text{ mol/s} \end{aligned}$$

$$\xi_2 = 4.76 \text{ mol/s} \Rightarrow \xi_1 = 46 \text{ mol/s} - 4.76 \text{ mol/s} = 41.24 \text{ mol/s}$$

$$\dot{n}_{\text{out}, \text{A}} = 41.24 \text{ mol/s}$$

$$\dot{n}_{\text{out}, \text{EA}} = 4.76 \text{ mol/s}$$

$$\dot{n}_{\text{out}, \text{H}} = 50.76 \text{ mol/s}$$

$$\xi_1 = 41.24 \text{ mol/s} \quad \xi_2 = 4.76 \text{ mol/s}$$

(Problem 2 Continued)

(b) Finish filling enthalpy values into the Input/Output Table below. Use a reference state of $T_{ref} = 25^\circ\text{C}$. (10 pts.)

$$\hat{H}_i = \Delta \hat{H}_i^0 + \Delta \hat{H}_{f,i}^0$$

specific enthalpy $25^\circ\text{C} \Rightarrow 298.15\text{K}$ $t = 0.298$

$$E \text{ at } 500\text{K}: \int_{0.298}^{0.5} C_p dt = \int_{0.298}^{0.5} 32.3 + 120t dt = \left[32.3t + \frac{120}{2}t^2 \right]_{0.298}^{0.5} \\ = 16.19 \times 1000 \text{ J/mol} = 16.19 \text{ kJ/mol}$$

$$E \text{ at } 800\text{K}: \int_{0.298}^{0.8} C_p dt = 49.29 \text{ kJ/mol}$$

$$A \text{ at } 800\text{K}: \int_{0.298}^{0.8} (30.7 + 88.2t) dt = 39.72 \text{ kJ/mol}$$

$$EA \text{ at } 800\text{K}: \int_{0.298}^{0.8} (60.9 + 191t) dt = 83.21 \text{ kJ/mol}$$

$$H \text{ at } 800\text{K}: \int_{0.298}^{0.8} (28.4 + 1.54t) dt = 14.68 \text{ kJ/mol}$$

$$\Delta \hat{H}_{f,i}^0$$

$$E: -234.95 \text{ kJ/mol}$$

$$A: -166.2 \text{ kJ/mol}$$

$$EA: -444.5 \text{ kJ/mol}$$

$$H: 0$$

Species	$\dot{n}_{in}$ (mol/s)	$\hat{H}_{in}$ (kJ/mol)	$\dot{n}_{out}$ (mol/s)	$\hat{H}_{out}$ (kJ/mol)
$\text{C}_2\text{H}_5\text{OH}$ E	120	-218.76	69.24	-185.66
CH_3CHO A	----	----	41.24	-126.48
$\text{CH}_3\text{COOC}_2\text{H}_5$ EA	----	----	4.76	-361.29
H_2 H	----	----	50.76	14.68

(Problem 2 Continued)

- (c) Calculate the heat rate in kJ/s for the reactor. Indicate clearly whether it is heat in or out of the reactor. (20 pts.)

$$\begin{aligned}\Delta H &= \sum_i (\dot{n}_i \hat{H}_i)_{\text{out}} - \sum_i (\dot{n}_i \hat{H}_i)_{\text{in}} \\ &= (69.24 \text{ mol/s}) (-185.66 \text{ kJ/mol}) + (41.24 \text{ mol/s}) (-126.48 \text{ kJ/mol}) \\ &\quad + (4.76 \text{ mol/s}) (-361.29 \text{ kJ/mol}) + (50.76) (14.68 \text{ kJ/mol}) \\ &\quad - (120 \text{ mol/s}) (-248.76 \text{ kJ/mol}) \\ &= \underline{7205.48 \text{ kJ/s}}\end{aligned}$$

$$\Delta H = \dot{W}_s + \dot{Q} \quad (\text{no work})$$

$$\dot{Q} = 7205.48 \text{ kJ/s} > 0.$$

∴ It is heat in to the reactor.

$$\dot{Q} = \underline{7205.48} \text{ kJ/s}$$

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